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The Canvas of Data Science



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Visual Programming: Place, Connect, Analyze



Place

Analytical steps are contained entirely within standalone widgets.



Connect

Workflows execute strictly from left to right. Orange never passes data backwards.



Analyze

Changes in one widget instantly propagate downstream through the entire canvas.



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The Paradigm Shift: Code vs. Canvas

Category	Traditional Data Science	Orange Visual Programming
Environment	Script-based, linear code blocks.	2D interactive canvas (Canvas).
Execution	Manual re-running of cells or script segments.	Live upstream and downstream data propagation.
Exploration	Static plots generated iteratively via libraries.	Interactive visualizations that output data subsets.
Accessibility	Steep learning curve, heavily syntax-dependent.	Highly accessible to domain experts without coding background.



Setting Up Your Workspace via Anaconda



The Environment: Create an isolated Python environment in Anaconda Navigator to keep dependencies clean.



The Channel: Add the *conda-forge* channel to access the most recent package bundles.



The Update: Regular updates ensure you have the latest bug fixes and experimental features.



Workflow Anatomy: Nodes, Ears, and Data Flow

Output "Ear"

Click and drag a link from the node's ear to instantly pass structured data to the next analytical block.



File



Data Table

Right-Click Canvas

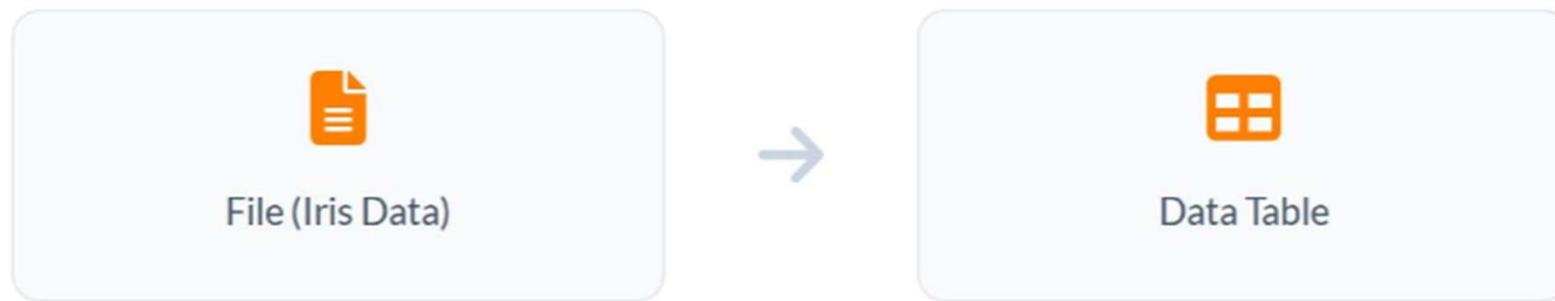
Right-click the canvas -> Type widget name -> Press Enter.

Drag & Drop

Drag a connection line from an output ear -> Release -> Select subsequent widget.



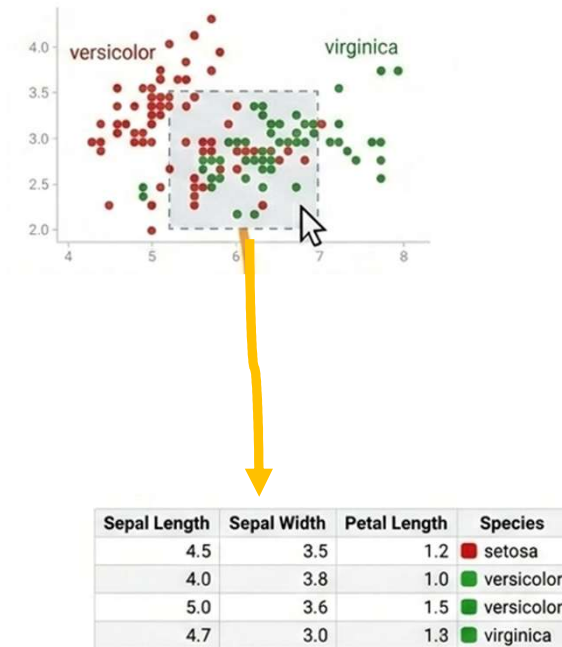
Your First Data Flow: The Iris Dataset



The data table is purely a viewer, the raw dataset remains fully intact and available at the source in the File widget

Interactive Exploratory

Visualizations in Orange act as dynamic filters. Selecting a cluster of points in a plot instantly propagates only that specific subset downstream for deeper inspection.



The Architecture of Predictive Modeling

Data

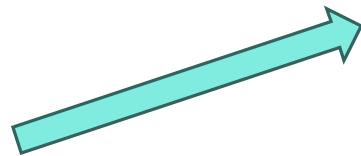
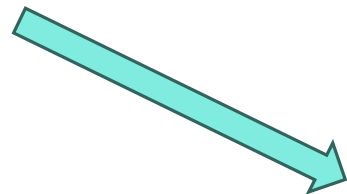
The dataset used for evaluation.

Learner(s)

Algorithms used for training (e.g., Logistic Regression, Random Forest).

Preprocessor (Optional)

Tools for normalization or feature selection.



Test and Score



Predictions

Once trained, the model passes here to predict outcomes on new, unseen data.

Scaling Complexity: Advanced Image Analytics

Import Images

Loads a folder of 82 images across 9 subfolders (daisies, lilies, roses, etc.). Subfolders automatically become class labels.

Image Viewer

A quick visual check to confirm all images loaded correctly.

Image Embedding

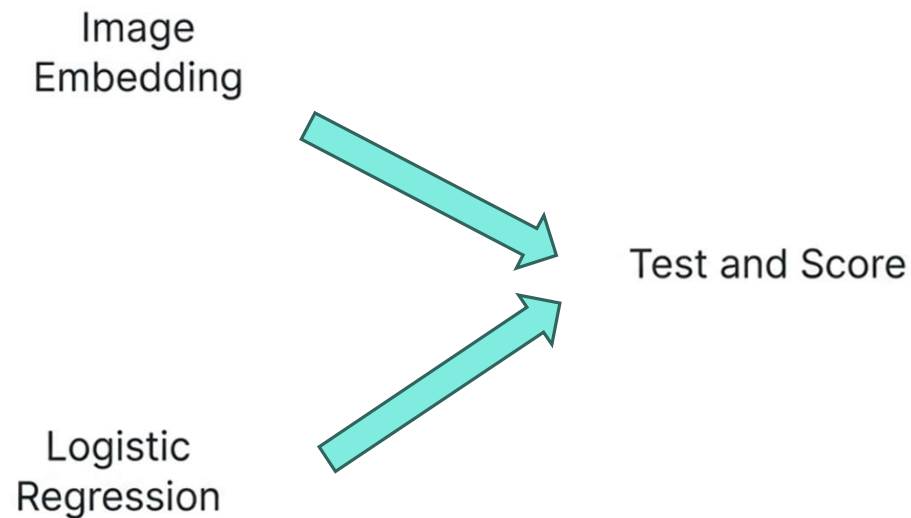
The heavy lifter. Passes images through deep networks, extracting 2,048 numerical features per image.

Data Table

The raw images successfully transformed into a 2,048-column predictive matrix.

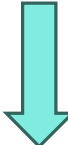


Evaluating the Image Model



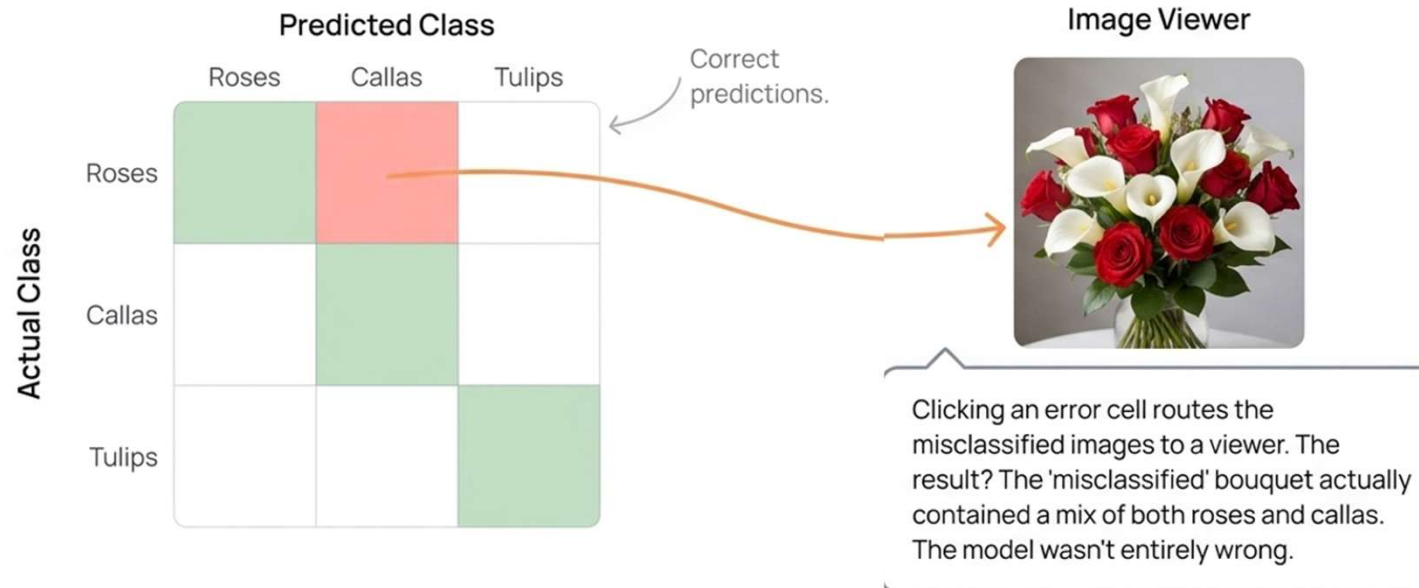
Model Performance Setup

- Method: 10-fold cross-validation.
- Result: High Area Under ROC Curve (AUC) and strong classification accuracy for the 9 flower classes.



The overall accuracy is high, but where exactly did the model fail?

Demystifying Misclassifications: The Confusion Matrix



A Complete Visualization Arsenal

Standard Visualizations

- Scatter Plots (Correlations)
- Box Plots (Basic statistics and spread)
- Heat Maps (Dataset overviews)

Exploratory Tools

- Interactive Subsetting (Routing selected points downstream)
- Silhouette Plots (Analyzing clustering results)
- Pythagorean Trees (Visualizing classification forests)

Intelligent Systems

- Rank Projections (Mathematical feature scoring)
- Score Plots (Finding best class separation automatically)



One Paradigm, Infinite Data Types

Tabular Data



Unstructured Images



Start Your Analysis



Single-Click Reporting

Orange captures your workflow history, intelligent visualizations, and model statistics directly into a comprehensive report with a single click.

Next Steps

- Install Orange via Anaconda Navigator.
- Load the pre-installed Iris dataset.
- Connect your first Scatter Plot.
- Explore advanced add-ons (Text, Network, Geo) via conda-forge.



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Thank You!



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